

Digital Twins for Next-Generation Wireless Communication Networks

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Physical Twin description

Modern wireless communication systems for *massive-scale terrestrial* or **non-terrestrial networks** rely on accurate characterization of signal generation, propagation and reception, user mobility, and the quantity or type of data to be communicated.

Digital Twin usages

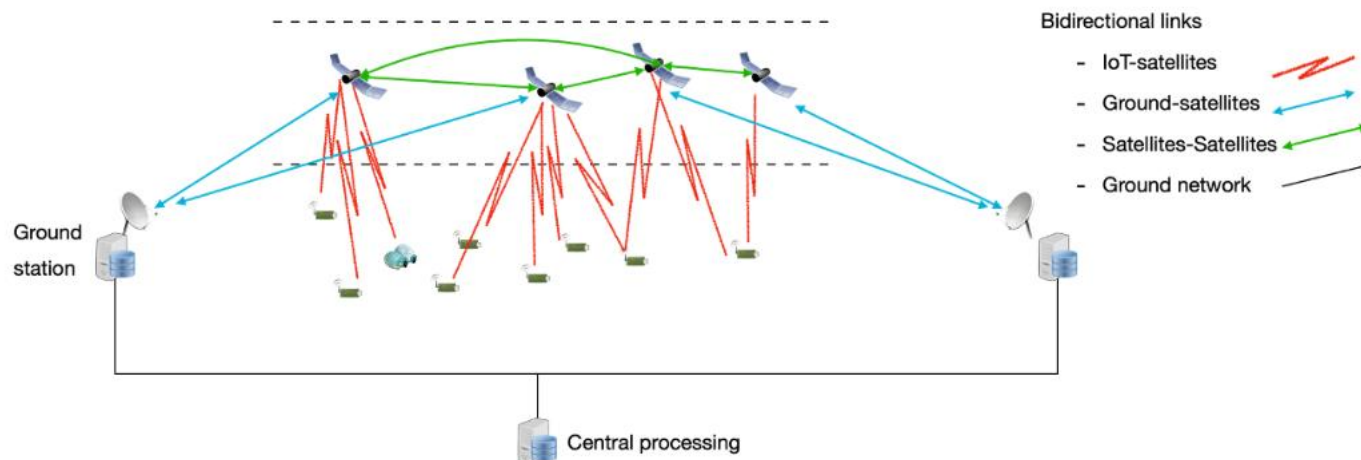
- Models and prediction of reliability and latency.
- Optimization of modulation, coding, and resource allocation.

Digital Twin Description

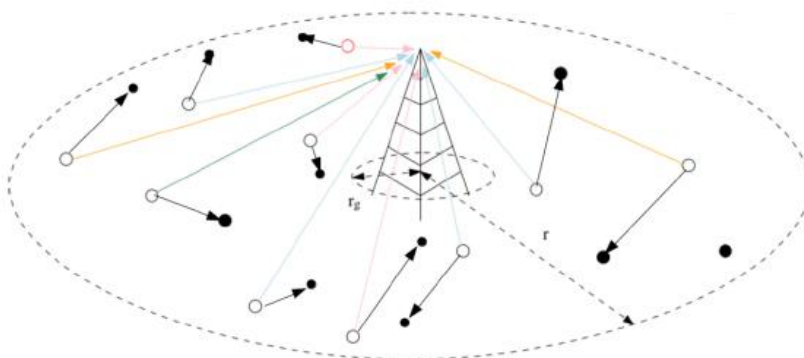
Digital twins for wireless communication systems aim to model and make predictions of reliability and latency based on models of the physical twin. This information forms the basis for optimization of modulation, coding, and resource allocation.

Engineering Digital Twin challenges

- Continual/online learning for physical twin models.
- Uncertainty quantification for reliability and latency.



Non-Terrestrial Communication Networks



Massive Multiple Access Communication Networks



SLICES-RI CorteXlab Testbed:

Reproducible experimentation for the physical layer.